

Mark schemes

Q1.

- (a) $6^2 = 4^2 + 5^2 - 2(4)(5) \cos \theta$
Use of the cosine rule

M1

$$\cos \theta = \frac{4^2 + 5^2 - 6^2}{2(4)(5)}$$

Rearrangement

m1

$$\cos \theta = \frac{5}{40} = \frac{1}{8}$$

CSO AG (be convinced)

A1

3

- (b) $\cos^2 \theta + \sin^2 \theta = 1$
Stated or used (PI)

M1

$$\sin^2 \theta = \frac{63}{64}$$

Or better

A1

$$\sin \theta = \frac{\sqrt{63}}{8} = \frac{\sqrt{9 \times 7}}{8} = \frac{3\sqrt{7}}{8}$$

AG (be convinced)

A1

3

- (c) Area of triangle = $0.5 \times 4 \times 5 \times \sin \theta$.

M1

$$= \frac{30\sqrt{7}}{8} \text{ cm}^2$$

.....

OE (Condone 9.92)

A1

Q2.

(a) (i) $2 (\operatorname{cosec}^2 x - 1) + 5 \operatorname{cosec} x = 10$

M1

$$2 \operatorname{cosec}^2 x - 2 + 5 \operatorname{cosec} x - 10 = 0$$

$$2 \operatorname{cosec}^2 x + 5 \operatorname{cosec} x - 12 = 0$$

AG

A1

2

(ii) $(2 \operatorname{cosec} x - 3)(\operatorname{cosec} x + 4) = 0$

Attempt to solve

M1

$$\operatorname{cosec} x = \frac{3}{2} \text{ or } -4$$

Condone answers with no method shown

A1

$$\sin x = \frac{2}{3} \text{ or } -\frac{1}{4}$$

AG

A1

3

(b) $(\theta - 0.1) = 0.73, 2.41, -0.25, -2.89$ AWRT

2 correct values, may be implied later

(41.8, 138.2, -165.5, -14.5)

B1

$\theta = 0.83, 2.51, -0.15, -2.79$ AWRT

2 correct answers

B1

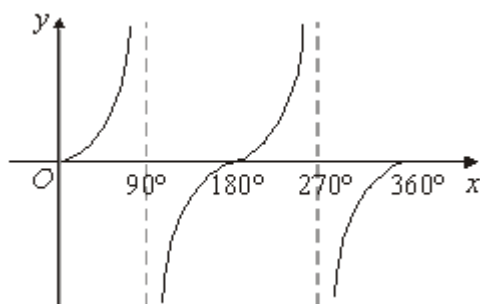
+ 2 correct answers and no extra within range

B1

3

Q3.

(a)



*Correct shape of branch from O to 90°
or correct shapes of branches from 90° - 360°*

M1

*Complete graph for $0^\circ \leq x \leq 360^\circ$
(Asymptotes not explicitly required but graphs should show "tendency")*

A1

Correct scaling on x -axis $0^\circ \leq x \leq 360^\circ$

A1

3

(b) 61° ;

For 61°

B1

241°

For 241° and no 'extras' in the interval $0^\circ \leq x \leq 360^\circ$

B1

2

(c) (i) $\sin \theta = -\cos \theta \Rightarrow \frac{\sin \theta}{\cos \theta} = -1$

$\Rightarrow \tan \theta = -1$

*AG; be convinced that the identity $\frac{\sin \theta}{\cos \theta} = \tan \theta$
is known and validly used*

B1

1

(ii) $\Rightarrow \tan(x - 20^\circ) = -1$

M1

$$x - 20^\circ = \tan^{-1}(-1)$$

m1

$$x - 20^\circ = 135^\circ, 315^\circ \dots$$

$$x = 155^\circ;$$

A1

$$335^\circ$$

Ft on (180 + "155") and no 'extras' in the given interval.

A1ft

4

(d) Translation

'Translation'/'translate(d)'

B1

$$\begin{bmatrix} 20 \\ 0 \end{bmatrix}$$

Accept equivalent in words provided linked to 'translation/move/shift' (Note: B0B1 is possible)

B1

2

(e) $f(x) = \tan 4x$

For tan 4x

B1

1

Q4.

(a) $2\operatorname{cosec}^2 x = 5(1 - \cot x)$

$$2 + 2 \cot^2 x = 5 - 5 \cot x$$

$$\text{use of } \operatorname{cosec}^2 x = 1 + \cot^2 x$$

M1

$$2 \cot^2 x + 5 \cot x - 3 = 0$$

AG

A1

2

[13]

(b) $(2 \cot x - 1)(\cot x + 3) = 0$
 or $2 + 5t - 3t^2 = 0$ Or in $\tan x$
 $(2 - t)(1 + 3t) = 0$

M1

$$\cot x = \frac{1}{2}, -3$$

$$\tan x = 2, -\frac{1}{3}$$

AG

A1

2

(c) $x = 1.1, -2.0$ AWRT

$x = -0.3, 2.8$ AWRT

Any 2 correct (In degrees): B0

Any 3 correct (In degrees): B1

4 correct (In degrees): B2

B1

B1

B1

[7]

Q5.

- (a) Stretch (I) in x -direction (II) scale
 Need(I) and one of (II),(III)

M1

factor 2 (III)

M0 if more than one transformation

A1

2

(b) $\tan^{-1} 3 = 1.2(49....) (= \alpha)$
 $\tan^{-1} 3$ [PI by 71.(56..)°]

M1

$$\left\{ \frac{1}{2}x \right\} = \pi + \alpha ;$$

Correct quadrant; condone degrees or mix

m1

$$\frac{1}{2}x = 1.249\dots; 4.3906\dots$$

$$x = 2.498\dots = 2.50 \text{ to 3 sf}$$

A1

$$x = 8.781\dots = 8.78 \text{ to 3 sf}$$

A1

*Condone 2.5 otherwise deduct **max** of 1 mark throughout this question from A marks if 'correct' rads. but to 2sf or final answers in degrees. (143°, 503°)*

As usual, accept greater accuracy answers. Ignore extra values outside the given interval (0 to 12.6). If > 2 values inside interval lose an A mark for each one.

NB M1m0A1A0 is possible

SC after M0 for error $\tan x = 6$;

Either $x = 1.40(5)$, $4.54(7)$, $7.68(8)$, $10.8(3)$ or $x = 80.5^\circ$, 260.5° , 440.5° , 620.5° SC
B1

(accept each rounded or truncated to 3 sf)

4

(c) $\cos \theta = 0$, $\sin \theta - 3 \cos \theta = 0$

Need both

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M1

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \text{ or } \tan \theta = 3$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \text{ seen/used}$$

M1

$$\cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2} = 1.57(07\dots)$$

Accept $\frac{\pi}{2}$

B1

or $\theta = \frac{3\pi}{2} = 4.71(23..)$

Accept $\frac{3\pi}{2}$

B1

$\tan \theta = 3 \Rightarrow \theta = 1.249...; 4.3906... = 1.25, 4.39 \text{ to } 3\text{sf}$

If not correct, ft on (b)

NB M0M1(B0B0)A1ft is possible

A1 ✓

$90^\circ; 270^\circ; 71.5(6)^\circ; 251.5(6)^\circ$

5

[11]

Q6.

(a) $\sec x = 5$
 $\cos x = 0.2$

M1

$x = 1.37, 4.91 \text{ AWRT}$

A1A1

3

(b) $\tan^2 x = 3 \sec x + 9$
 $\sec^2 x - 1 = 3 \sec x + 9$

for using $\sec^2 x = 1 + \tan^2 x$ OE

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$\sec^2 x - 3 \sec x - 10 = 0$

AG

A1

2

(c) $(\sec x - 5)(\sec x + 2) = 0$
or use of formula (attempt)

M1

$\sec x = 5, -2$

A1

$$\cos x = 0.2, -0.5$$

$$x = 1.37, 4.91$$

any 2 correct or ft their 2 answers in (a)

B1F

$$2.09, 4.19$$

all 4 correct, no extras

A1

4

[9]

Q7.

(a) $x = \frac{1}{2} \quad y = \frac{\pi}{2} \quad \left(\text{or } 1.57, \sin^{-1} 1 \right)$

ignore 90°

B1

1

(b) (i) $y = \sin^{-1} 2x$
 $\sin y = 2x$ and $\frac{1}{2} \sin y = x$
AG (be convinced)

B1

1

(ii) $\frac{dx}{dy} = \frac{1}{2} \cos y$

B1

1

(c) $\frac{dy}{dx} = \frac{2}{\cos y}$

$\frac{k}{\cos y}$
M1 for

M1A1

$\sin y = 2x$ and $\sin^2 + \cos^2 = 1$

use of to get $\cos y$ or $\cos^2 y$

M1

$$\cos y = \sqrt{1 - 4x^2}$$

$$\frac{dy}{dx} = \frac{2}{\sqrt{1 - 4x^2}}$$

AG; condone omission of proof of sign

A1

4

[7]

Q8.

(a) (i) $4 \frac{\sin \theta}{\cos \theta} \sin \theta = 15$

$$\Rightarrow 4 \sin^2 \theta = 15 \cos \theta$$

AG Be convinced

B1

1

(ii) $\sin^2 \theta + \cos^2 \theta = 1$

OE seen

M1

$$4(1 - \cos^2 \theta) = 15 \cos \theta$$

$$4 \cos^2 \theta + 15 \cos \theta - 4 = 0$$

AG Be convinced

A1

2

(b) (i) $(4c - 1)(c + 4) = 0$

Factorisation or formula or completion of square

M1

$$c = -4, c = \frac{1}{4}$$

Both values

A1

2

(ii) Since $-1 \leq \cos \theta \leq 1$, the only possible value for $\cos \theta$ is $\frac{1}{4}$

AG convincingly explained

(Condone strict inequalities)

ft provided candidates answers for c

are $\frac{1}{4}$ and a value k such that $k > 1$ or $k < -1$

E1ft

1

(iii) $\theta = 75.5^\circ$

B1

$\theta = 284.5^\circ$

Ft on $[360 - c's 75.5^\circ]$ as only other solution in the given interval

B1ft

2

(c) $\Rightarrow \cos 4x = \frac{1}{4}$

Links with previous parts. P1

M1

$x = 19^\circ, 71^\circ$

ft on (iii)/4....(only ft if 2 answers in given range).

A1ft

2

[10]

Q9.

(a) $\tan^2 x = \sec x + 11$

Or attempt to form quadratic in $\cos^2$

$\sec^2 x - 1 = \sec x + 11$

$\tan^2 x = \sec^2 x - 1$

M1

$\sec^2 x - \sec x - 12 = 0$

AG

A1

2

(b) $(\sec x - 4)(\sec x + 3) = 0$

attempt at solving quadratic

M1

$$\sec x = 4, -3$$

A1F

$$\therefore \cos x = \frac{1}{4}, -\frac{1}{3}$$

$$AG; (A0 \text{ if no use of } \cos x = \frac{1}{\sec x})$$

A1

3

(c) $x = 76^\circ, 284^\circ$

2 correct

B1

$$x = 109^\circ, 251^\circ \text{ (or better)}$$

other answers (–1 each extra in range)

B1,B1

3

If radians $x = 1.32, 4.97$
 $1.91, 4.37$

B1 any 2 correct

B1 other 2 correct

[8]

Q10.

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(a) $x = ut \cos \alpha$

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M1

$$t = \frac{x}{u \cos \alpha}$$

A1

$$y = -\frac{1}{2}gt^2 + ut \sin \alpha$$

Must have correct signs

M1

$$y = -\frac{1}{2}g\left(\frac{x}{u \cos \alpha}\right)^2 + u\left(\frac{x}{u \cos \alpha}\right) \sin \alpha$$

M1

$$y = -\frac{gx^2}{2u^2 \cos^2 \alpha} + \frac{x \sin \alpha}{\cos \alpha}$$

$$y = -\frac{gx^2}{2u^2} (1 + \tan^2 \alpha) + x \tan \alpha$$

A1

$$k = -\frac{10(2k)^2}{2u^2} (1 + \tan^2 \alpha) + 2k \tan \alpha$$

M1

$$u^2 = -20k (1 + \tan^2 \alpha) + 2u^2 \tan \alpha$$

$$20k \tan^2 \alpha - 2u^2 \tan \alpha + u^2 + 20k = 0$$

AG

A1

7

(b) Pass through $P \Rightarrow$ Discriminant ≥ 0

$$(-2u^2)^2 - 4(20k)(u^2 + 20k) \geq 0$$

OE must be seen

M1A1

$$4u^4 - 80ku^2 - 1600k^2 \geq 0$$

$$u^4 - 20ku^2 - 400k^2 \geq 0$$

AG

A1

3

[10]

Q11.

(a) $x = 40 \cos \theta . t$

M1

$$y = -\frac{1}{2}(10)t^2 + 40 \sin \theta . t$$

M1 A1

$$y = -\frac{1}{2}(10)\left(\frac{x}{40 \cos \theta}\right)^2 + 40 \sin \theta . \left(\frac{x}{40 \cos \theta}\right)$$

Dependent on both M1s

m1

$$y = -\frac{x^2}{320 \cos^2 \theta} + x \tan \theta$$

$$320y = -x^2(1 + \tan^2 \theta) + 320x \tan \theta$$

m1

$$x^2 \tan^2 \theta - 320x \tan \theta + (x^2 + 320y) = 0$$

Answer Given (Condone missing brackets)

A1

6

(b) (i) $150^2 \tan^2 \theta - 320(150)\tan \theta + (150^2 + 320 \times 8) = 0$

M1

$$1125 \tan^2 \theta - 2400 \tan \theta + 1253 = 0$$

Correct quadratic

A1

$$\tan \theta = \frac{2400 \pm \sqrt{2400^2 - 4(1125)(1253)}}{2(1125)}$$

m1

$$\tan \theta = 1.22, \quad 0.912$$

PI

A1F

$$\theta = 50.7^\circ, \quad 42.4^\circ$$

A1F

5

(ii) $\theta = 42.4^\circ$

For the smaller angle

B1F

$$t = \frac{150}{40 \cos \theta} \text{ and } \cos 42.4 > \cos 50.7$$

OE

E1

2

[13]

Q12.

(a) (i) $x = 80 \cos \theta \cdot t$

B1

$$t = \frac{x}{80 \cos \theta}$$

B1

$$y = 80 \sin \theta \cdot t - \frac{1}{2} g t^2$$

B1

$$y = 80 \sin \theta \frac{x}{80 \cos \theta} - \frac{1}{2} g \left(\frac{x}{80 \cos \theta} \right)^2$$

M1

$$y = x \tan \theta - \frac{g x^2}{12800} (1 + \tan^2 \theta)$$

Answer given

A1

5

$$(ii) \quad -20 = 400 \tan \theta - \frac{9.8 \times 400^2}{12800} (1 + \tan^2 \theta)$$

Condone + 20

M1

$$122.5 \tan^2 \theta - 400 \tan \theta + 102.5 = 0$$

$$49 \tan^2 \theta - 160 \tan \theta + 41 = 0$$

Answer given

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A1

2

$$(b) \quad (i) \quad \tan \theta = \frac{160 \pm \sqrt{25600 - 4(49)(41)}}{2 \times 49}$$

M1

$$= 2.9850, 0.2803$$

PI

A1

$$\theta = 71.5^\circ, 15.7^\circ$$

A1F

3

- (ii) For the shortest time
 $400 = 80 \cos 15.7^\circ \cdot t$

M1

$$t = 5.19$$

A1F

2

- (c) • The projectile is a particle
 • The air resistance is negligible

E1

1

[13]

Q13.

- (a) $5 = 10 \cos \alpha \cdot t$

M1

$$t = \frac{5}{10 \cos \alpha}$$

A1

$$1 = -\frac{1}{2}(9.8)t^2 + 10 \sin \alpha t$$

M1A1

$$1 = -\frac{1}{2}(9.8) \frac{25}{100 \cos^2 \alpha} + 10 \sin \alpha \frac{5}{10 \cos \alpha}$$

Dependent on both M1s

M1

$$1 = -\frac{1}{2}(9.8) \frac{25}{100} (1 + \tan^2 \alpha) + 10 \sin \alpha \frac{5}{10 \cos \alpha}$$

A1

$$49 \tan^2 \alpha - 200 \tan \alpha + 89 = 0$$

Answer given

A1

7

(b) $\tan \alpha = \frac{200 \pm \sqrt{40000 - 4(49)(89)}}{2 \times 49}$

M1

$= 3.57, 0.508$

AWRT

A1

$\alpha = 74.4^\circ, 26.9^\circ$

A1F

3

- (c) (i) $10 \cos 26.9^\circ = 8.92$ (or 8.91) > 8
 $\Rightarrow$ The can will be knocked off the wall
Both values checked

M1

$10 \cos 74.4^\circ = 2.69 < 8$

*Acc. of both results
 Correct conclusions*

A1F

$\Rightarrow$ The can will not be knocked off the wall

E1

ALTERNATIVE

The can will be knocked off the wall if

$10 \cos \alpha > 8$

$\cos \alpha > 0.8$

$\alpha < 36.9^\circ$

So, for $\alpha = 26.9^\circ$ the can will be knocked off

and for $\alpha = 74.4^\circ$, the can will not be knocked off

M1A1

E1

3

(ii) $x = ut$

$t = \frac{5}{10 \cos 26.9^\circ}$

$v = 10 \sin 26.9^\circ - 9.8 \left(\frac{5}{10 \cos 26.9^\circ} \right)$

Any correct use of equations

M1

$v = -0.970$

A1F

$$\tan \theta = \frac{-0.970}{8.92}$$

M1

$$\theta = -6.2^\circ$$

At an angle of depression of 6.2°
AWRT 6°

A1F

4

[17]

Q14.

(a) $y = ut \sin \alpha - \frac{1}{2}gt^2$

M1A1

$$x = ut \cos \alpha$$

M1

$$t = \frac{x}{u \cos \alpha}$$

A1

$$y = u \left(\frac{x}{u \cos \alpha} \right) \sin \alpha - \frac{1}{2}g \left(\frac{x}{u \cos \alpha} \right)^2$$

M1

$$y = x \tan \alpha - \frac{gx^2}{u^2 \cos^2 \alpha}$$

$$y = x \tan \alpha - \frac{gx^2}{2u^2} (1 + \tan^2 \alpha)$$

Answer given

A1

6

(b) (i) $1 = R \tan \alpha - \frac{10R^2}{2u^2} (1 + \tan^2 \alpha)$

M1

$$5R^2 \tan^2 \alpha - u^2 R \tan \alpha + 5R^2 + u^2 = 0$$

Answer given

A1

2

(ii) For real solutions of the quadratic:

$$u^4 R^2 - 20R^2(5R^2 + u^2) \geq 0$$

M1

$$R^2 \leq \frac{u^4 - 20u^2}{100}$$

$$R^2 \leq \frac{u^2(u^2 - 20)}{100}$$

Answer given

A1

2

(iii)

$$5^2 \leq \frac{u^2(u^2 - 20)}{100}$$

$$u^4 - 20u^2 - 2500 \geq 0$$

Condone equation

M1

$$u_{\min}^2 = 61.0 \quad (\text{or } 10 + \sqrt{2600})$$

A1

$$u_{\min} = 7.81$$

3 sf required

A1F

3

[13]

Q15.

(a)

$$y = -\frac{1}{2}gt^2 + 20\sin 30t$$

M1A1

$$x = 20 \cos 30t$$

M1

$$t = \frac{x}{20 \cos 30}$$

A1

$$y = -\frac{1}{2}g \frac{x^2}{400 \cos^2 30} + 20 \sin 30 \frac{x}{20 \cos 30}$$

M1

$$y = x \tan 30 - \frac{gx^2}{800 \cos^2 30^\circ}$$

AG

A1

6

(b) $2.5 = x \tan 30 - \frac{9.8x^2}{800 \cos^2 30}$

$$9.8x^2 - 346x + 1500 = 0$$

substituting and tidying up

M1A1

$$x = \frac{346 \pm \sqrt{119716 - 58800}}{19.6}$$

M1

=30.3 (or 30.2) & 5.06 (or 5.05)

answer: 30.3m (or 30.2m)

at least 3 s.f. required

A1F

4

(c) no air resistance,

B1

the ball is a particle

B1

2

etc.

[12]

Q16.

(a) $x = 40t \cos \alpha$

M1

$$y = 40t \sin \alpha - \frac{1}{2} g t^2$$

M1A1

$$t = \frac{x}{40 \cos \alpha}$$

$$y = 40 \left(\frac{x}{40 \cos \alpha} \right) \sin \alpha - \frac{1}{2} g \left(\frac{x}{40 \cos \alpha} \right)^2$$

m1

$$y = x \tan \alpha - \frac{g x^2}{3200 \cos^2 \alpha}$$

$$y = x \tan \alpha - \frac{g x^2}{3200} (1 + \tan^2 \alpha)$$

Answer given

A1

5

(b) $4 = 100 \tan \alpha - \frac{9.8 \times 100^2}{3200} (1 + \tan^2 \alpha)$

M1

$$245 \tan^2 \alpha - 800 \tan \alpha + 277 = 0$$

M1A1

$$\tan \alpha = \frac{400 \pm \sqrt{(-400)^2 - (245)(277)}}{245}$$

Or equivalent

A1F

M1

$\alpha = 71^\circ, 21^\circ$ (22° acceptable)
(or 1.24 rad., 0.375 rad.)

A.T.W.R.T

(or equivalent in radians)
Must be positive

A1F

6

(c) The ball is a particle

E1

No air resistance, etc

E1

2

[13]



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